PORTI LAB 6

Lab 06: Azure Monitor Agent (AMA) – Enterprise Monitoring & Security Telemetry Architecture

Region: West US 2
Resource Group: `AZ500LAB131415`
Objective: Design and implement a centralized telemetry architecture using Azure Monitor Agent (AMA) and Data Collection Rules (DCRs) to provide security visibility, performance monitoring, and operational intelligence across Azure virtual machines handling sensitive workloads.

Architectural Intent

In production cloud environments, virtual machines must be continuously observable. Without centralized telemetry, detection, investigation, and performance optimization become reactive and fragmented.

This lab implements a scalable monitoring architecture that:

- Centralizes performance telemetry
- Enables log-based investigation using KQL
- Establishes a foundation for SIEM integration
- Reduces operational blind spots

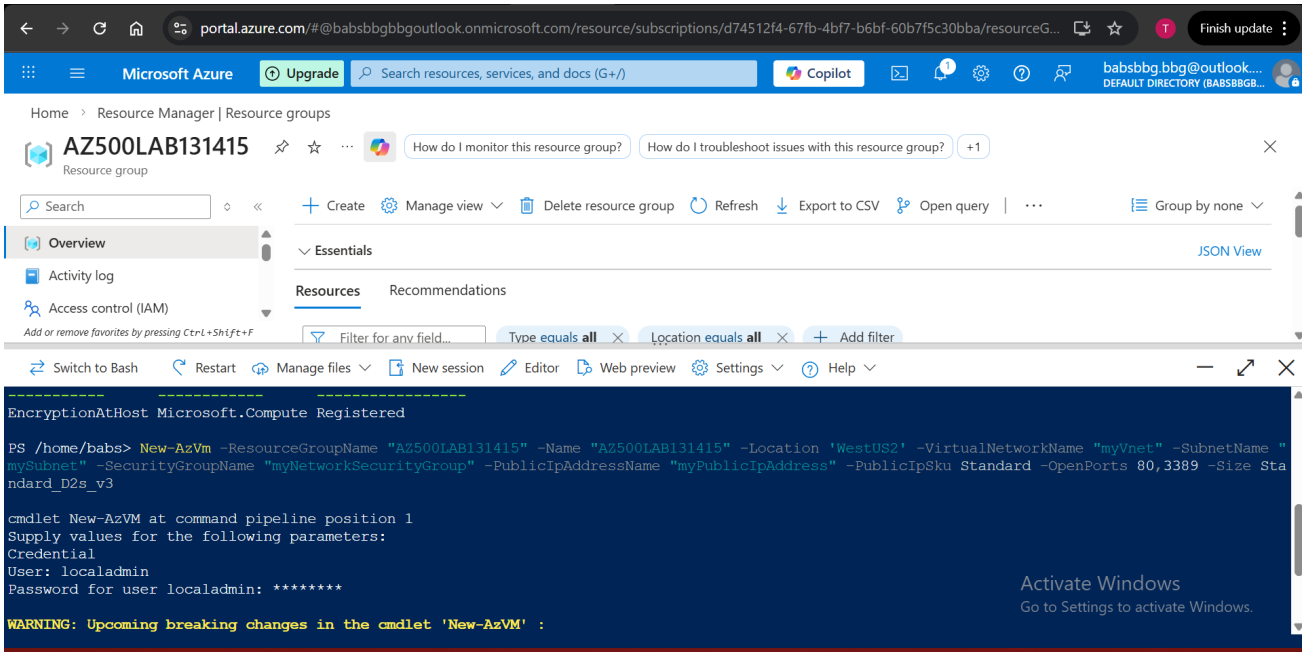
Exercise 1: Monitoring Foundation Deployment

Step 1 — Create Monitoring Resource Group

Created a dedicated resource group to isolate monitoring infrastructure.

- **Name:** `AZ500LAB131415`

- **Region:** West US 2



Azure Portal screenshot showing RG provisioning

Step 2 — Deploy Monitored Virtual Machine

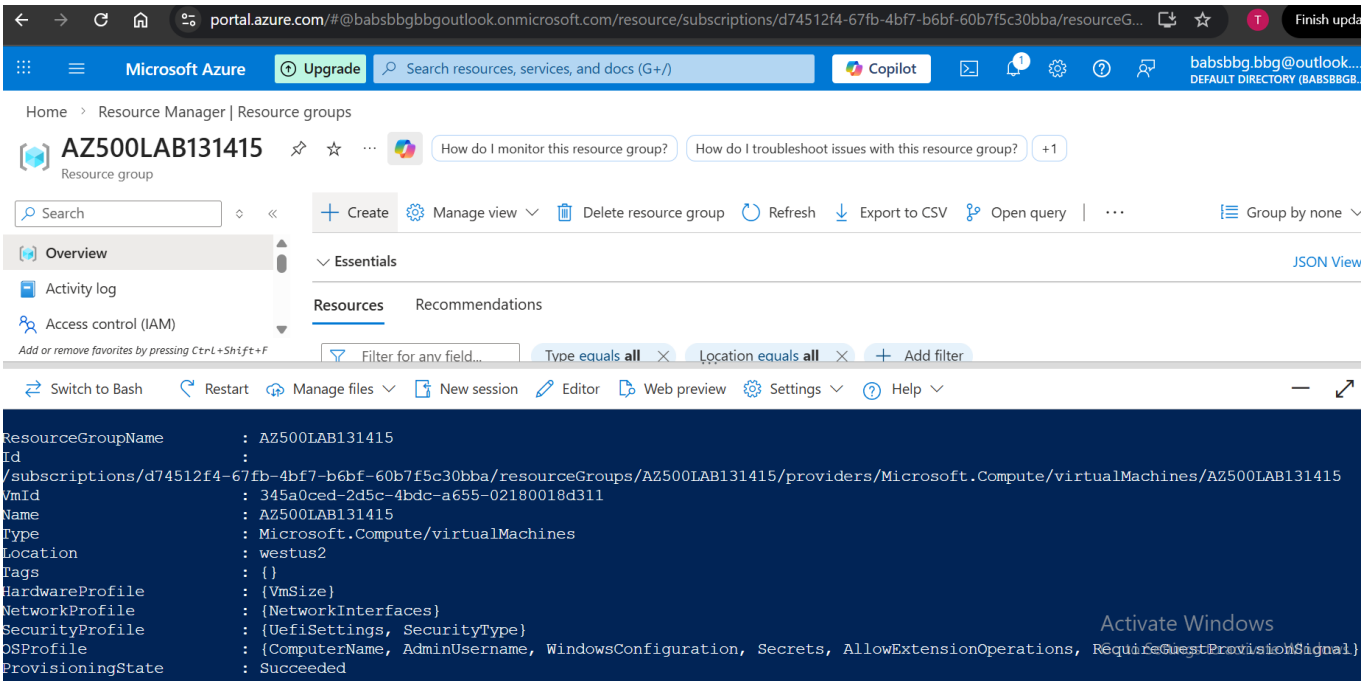
Provisioned Windows Server VM to act as monitored workload.

- VM Name: `AZ500LAB131415`
- OS: Windows Server 2022
- Region: West US 2
- RDP enabled for validation

Operational constraints encountered:

- Regional SKU limitations
- Core quota restrictions

Deployment parameters adjusted to align with availability.



VM overview screenshot showing status, region, and IP configuration

Exercise 2: Centralized Log Repository

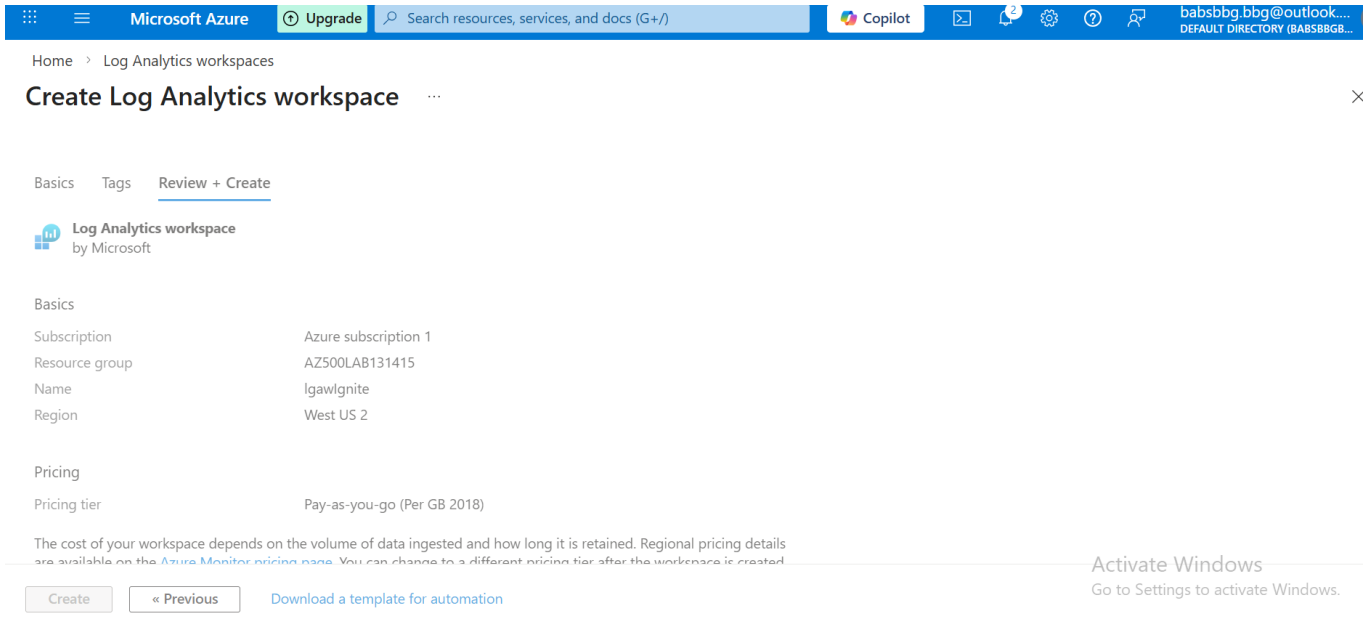
Step 3 — Create Log Analytics Workspace

Provisioned centralized telemetry repository.

- Workspace Name: `lgawIgnite`
- Region: West US 2

Purpose:

- Central storage of VM telemetry
- KQL-based investigation
- Future SIEM integration readiness



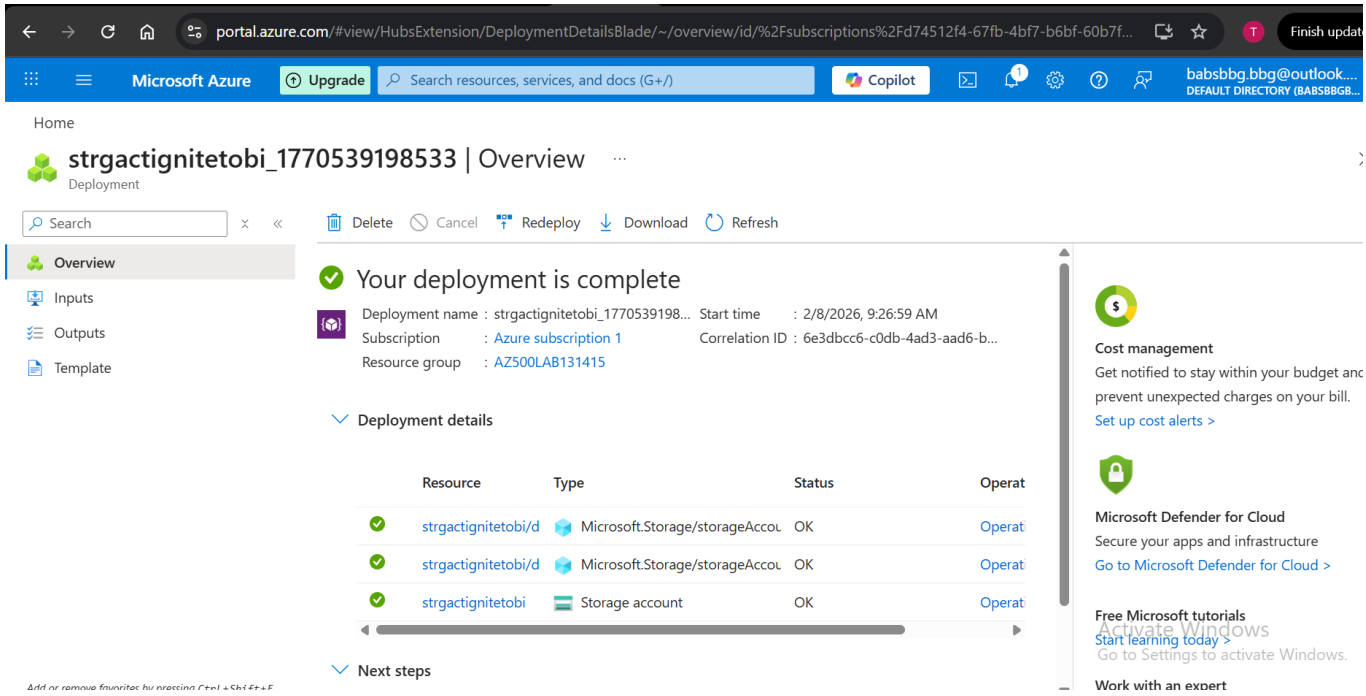
LAW overview screenshot showing region and resource group

Step 4 — Deploy Supporting Storage Account

Provisioned storage account for monitoring support.

- Name: `strgactignitetobi`
- Redundancy: LRS
- Performance: Standard

Note: Azure storage accounts require globally unique naming.



Storage Account Overview

Exercise 3: Azure Monitor Agent & Data Collection Architecture

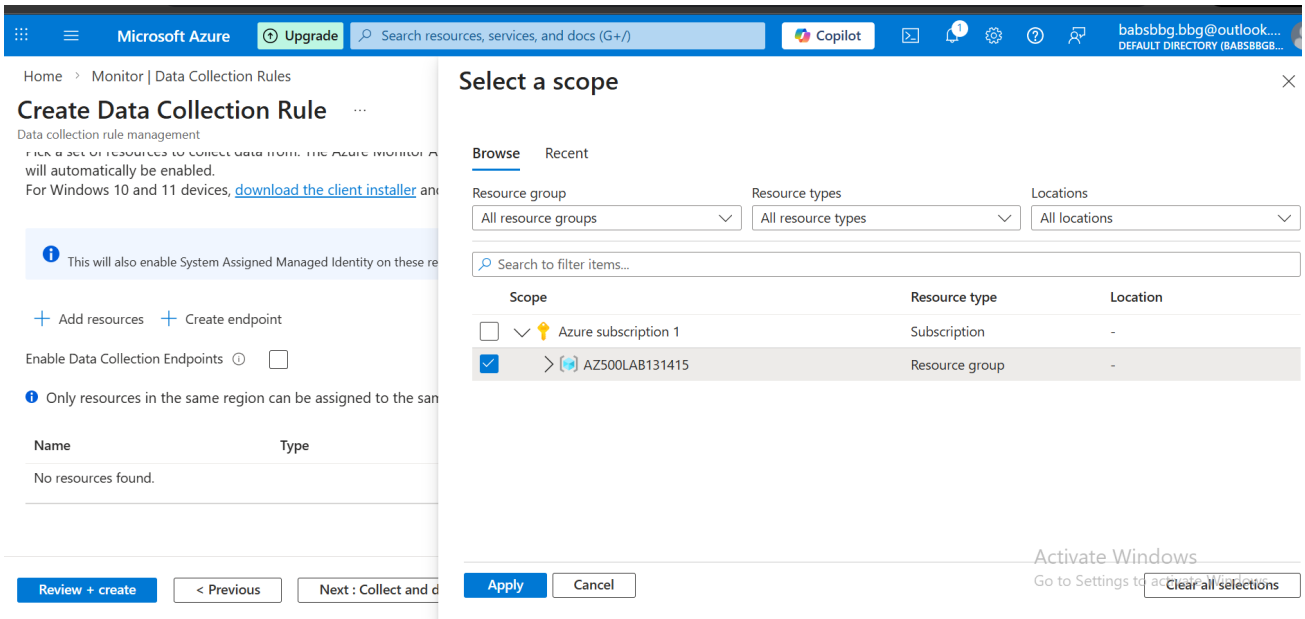
Step 5 — Create Data Collection Rule (DCR)

Configured policy-driven telemetry routing.

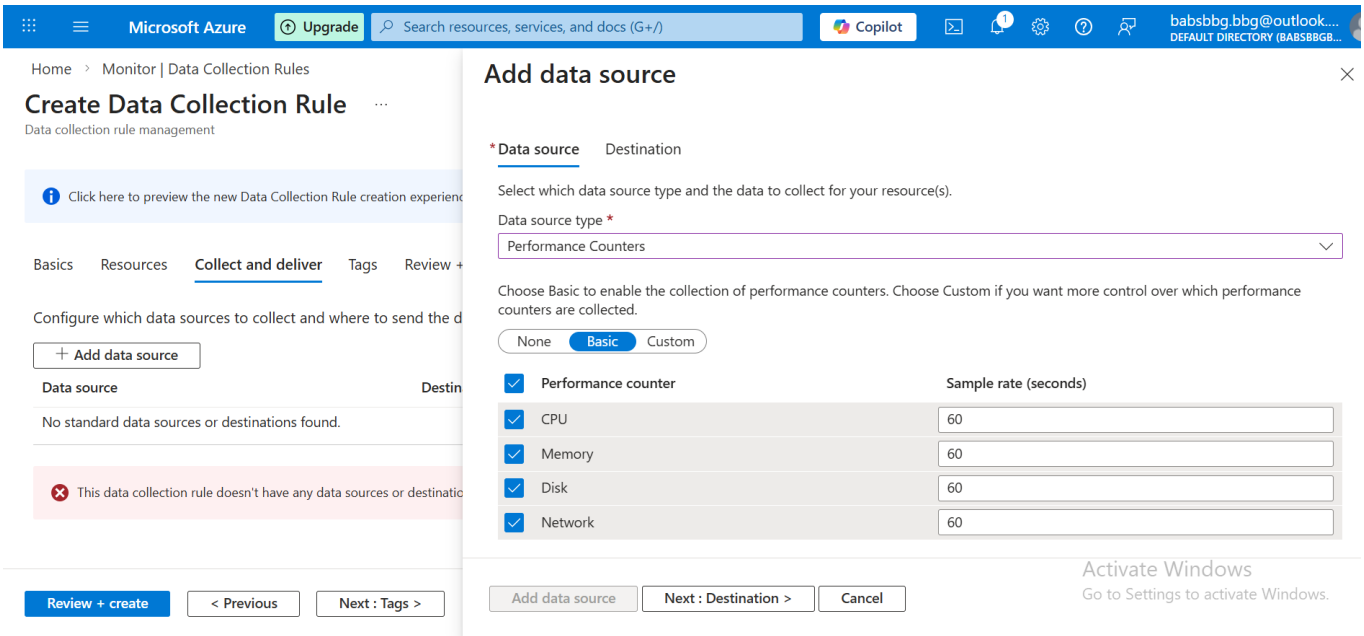
DCR Name: `DCR1`
Platform: Windows
Data Source: Performance Counters (CPU, Memory, Disk, Network @ 60s)
Destination: Log Analytics Workspace (`lgawIgnite`)

Architectural impact:

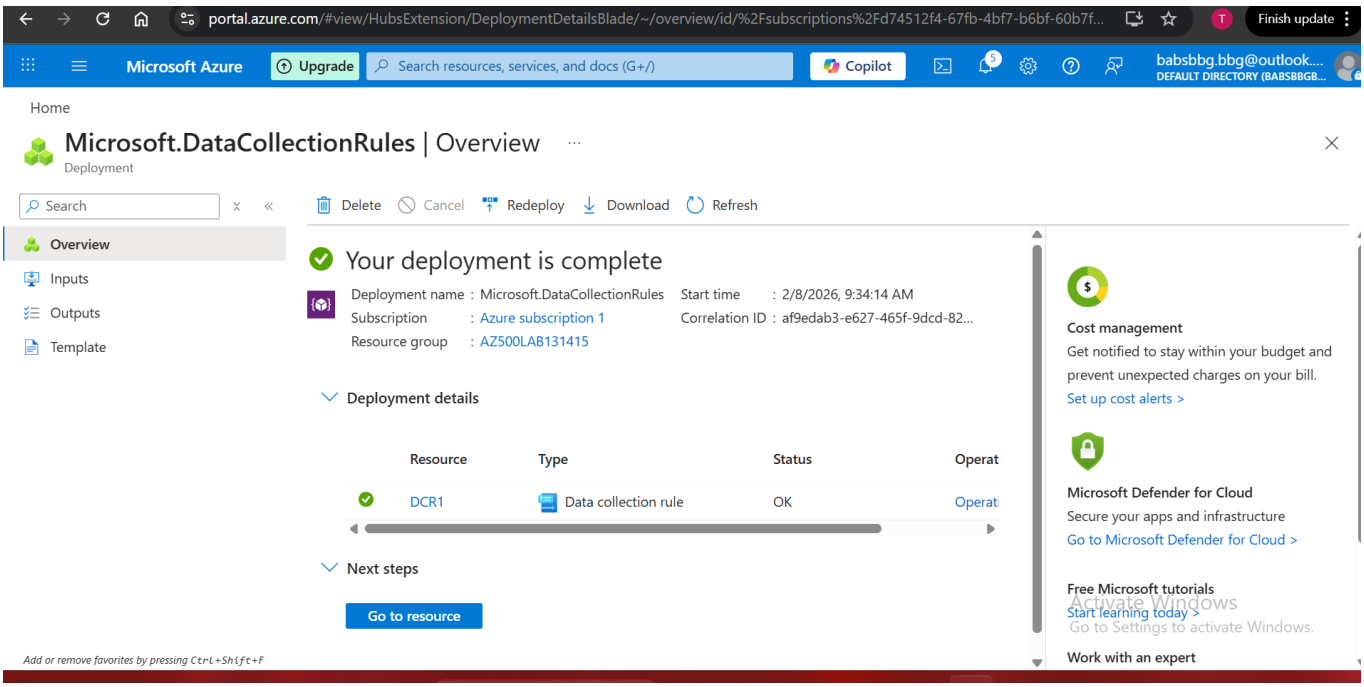
- Enables scalable monitoring onboarding
- Separates collection policy from workload configuration
- Standardizes telemetry pipeline



Data Collection Rule Overview (DCR1)



DCR Data Sources Configuration (Performance Counters)



DCR Resource Association

Exercise 4: Telemetry Validation & Verification

Step 6 — Generate Runtime Activity

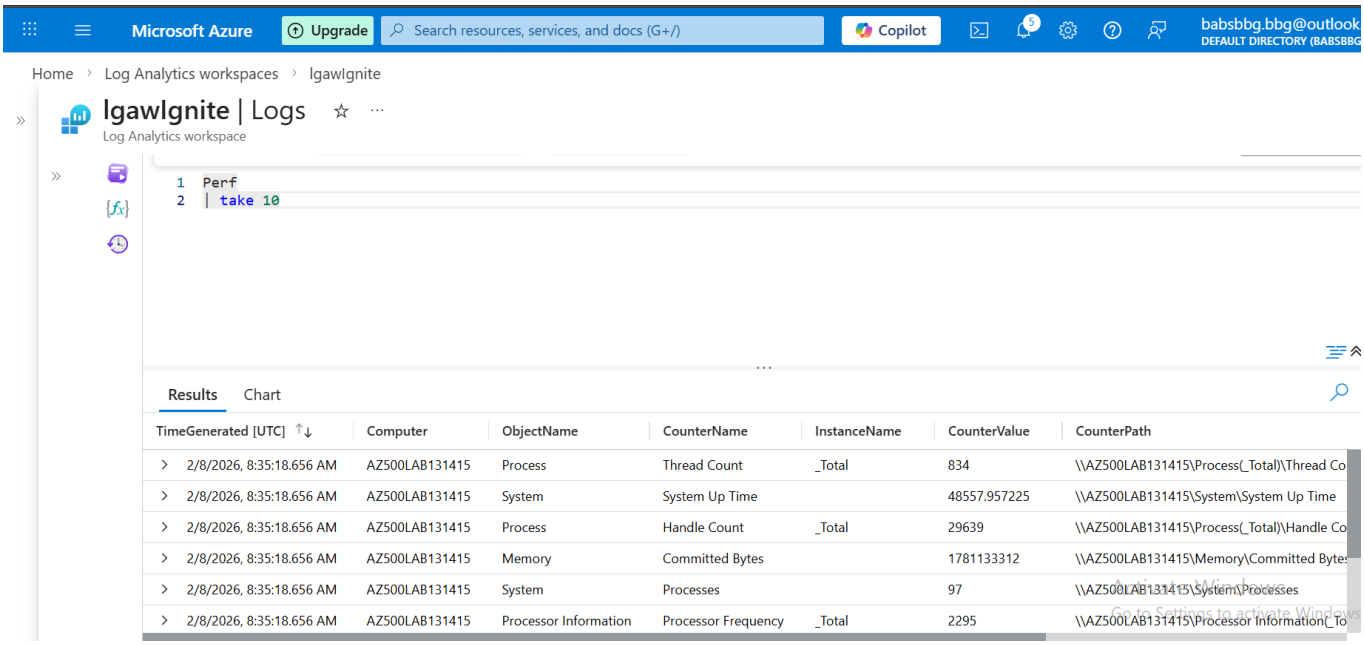
Connected via RDP and generated system activity to trigger performance telemetry.

Step 7 — Validate Ingestion Using KQL

Executed the following query in Log Analytics Workspace:

```
Perf | take 10
```

Initial ingestion delay observed (expected behavior).
After activity generation and time-range adjustment, performance telemetry successfully appeared.



KQL Query Execution (Perf Table Results Visible)

Enterprise Telemetry Flow

```
Azure Virtual Machine ↓ Azure Monitor Agent (AMA) ↓ Data Collection Rule (Policy Engine) ↓
Log Analytics Workspace ↓ Security Analytics & Investigation
```

Azure Resource Visualizer Diagram (Full Monitoring Architecture)

Security Engineering Outcome

This implementation achieved:

- Centralized telemetry aggregation
- 60-second performance monitoring
- Policy-driven monitoring enforcement
- Scalable onboarding model
- Verification of log pipeline integrity
- Reduced observability blind spots

Architectural Maturity Demonstrated

This lab reflects:

- Cloud-native monitoring design
- Security-first infrastructure mindset
- Policy-based telemetry routing

- Operational constraint handling
- Verification-driven validation

Final Assessment

The Azure Monitor Agent and Data Collection Rule architecture was successfully deployed and validated. The monitored workload is now centrally observable, enabling performance analytics, anomaly detection, and future SIEM integration.

This lab demonstrates practical implementation of enterprise-grade monitoring architecture within Azure.